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Hiring Manager

Terranova

Berkeley, CA

Dear Hiring Manager,

I am writing to apply for the Robotics Controls Engineer position at Terranova. As a robotics M.Eng. candidate at UC Berkeley with hands-on experience designing PID and MPC controllers for multi-robot platforms on STM32 and ROS2, I can contribute directly to the classical controls backbone that keeps Atlas, Prometheus, and Vulcan operating safely in the field.

My strongest alignment with Terranova's needs lies in real-time controller development. On the L.I.F.T. project, I built a ROS2-based robotic system coupling an MPC controller with a physics simulator for obstacle avoidance and full trajectory completion. To close the loop, I integrated an OptiTrack motion capture system delivering sub-millimeter accuracy at under 10 ms latency for sub-centimeter trajectory tracking during iterative controller tuning. This pipeline—model, controller, real-time sensor feedback—mirrors the simulation-to-field workflow Terranova's fleet demands under GPS-degraded conditions.

Beyond controls software, I bring the embedded-hardware depth Terranova values as a "huge bonus." During the ROBOCON competition, I wrote embedded C firmware for STM32F407 managing CAN motor buses, BLE communication, and triple-PID chassis control on an omnidirectional robot, achieving ± 2 mm tracking and 40% directional stability gains. On a separate crane project, I consolidated five boards into a single 4-layer PCB in KiCad, cutting footprint by 40%—the vertical integration a seed-stage hardware company needs when iterating on novel platforms. That Terranova can lift an acre of San Rafael by one foot per day at a tenth of the \$900M seawall cost motivates me to build at that intensity.

I want to be transparent: I do not yet have production ML/RL experience with PyTorch or JAX. My controls and embedded foundation, however, positions me to extend into learned-control methods quickly—and Terranova's mission of reshaping flood-vulnerable land is exactly the problem worth learning fast for. I would welcome the chance to discuss how my skills can serve your fleet's autonomy goals.

Sincerely,

Felix Hu